BUSINESS INTELLIGENCE LAB

EXPERIMENT 6

Dashboard and Scorecard Development using Microsoft Power BI

Report

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1. AIM

To design and develop an interactive Executive Dashboard and Business Scorecard using Microsoft Power BI for monitoring organizational performance, supporting business decision-making, and communicating key performance indicators (KPIs) effectively.

2. TOOLS / SOFTWARE REQUIRED

1. Microsoft Power BI Desktop
2. Global Superstore Dataset.xlsx
3. Power Query Editor
4. DAX (Data Analysis Expressions)

3. THEORY

Dashboard: A Dashboard is a visual interface that presents important business metrics, trends, and performance indicators in a consolidated view. Dashboards help executives and managers monitor organizational performance and make informed decisions.

Scorecard: A Scorecard is a management tool used to compare actual performance against predefined targets. Scorecards help organizations track progress toward strategic objectives using Key Performance Indicators (KPIs).

Microsoft Power BI: Power BI provides a rich set of visualization tools, KPI cards, charts, slicers, filters, and interactive features that allow users to build professional dashboards and scorecards.

4. DATASET DESCRIPTION

Dataset Name: Global Superstore Dataset

Dataset Source: Kaggle

Dataset Link: https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset

The Global Superstore dataset contains information related to orders, sales, profits, customers, regions, product categories, and segments. It is used to analyze business performance and generate meaningful insights through data visualization.

5. BUSINESS SCENARIO

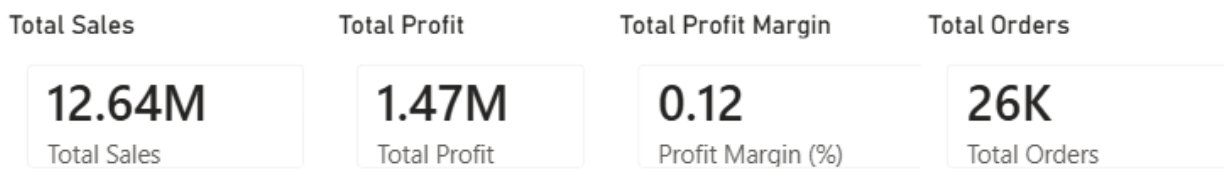
You are working as a Business Intelligence Analyst for Global Superstore. The executive management team requires a one-page dashboard and performance scorecard to monitor sales, profit, customer growth, and operational performance.

The task is to design a professional dashboard and provide recommendations based on the analytical findings.

6. IMPLEMENTATION / PROCEDURE

Part A: Dashboard Planning

1. Opened the existing Power BI project.
2. Identified key business metrics required by management.
3. Selected suitable visualizations for presenting performance indicators.
4. Planned the dashboard layout before implementation.



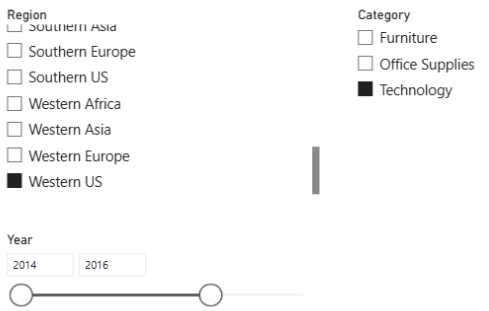
Part B: Create KPI Dashboard

1. Created KPI Cards for Total Sales, Total Profit, Total Orders, and Profit Margin (%).
2. Created visualizations for:
    - Sales by Region
    - Sales by Category
    - Monthly Sales Trend
    - Top 10 Customers by Sales
    - Profit by Segment
3. Formatted the visuals for readability and consistency.



Part C: Add Interactive Features

1. Added slicers for Region, Category, and Year.
2. Configured the slicers to interact dynamically with dashboard visuals.
3. Tested the dashboard interactivity and usability.



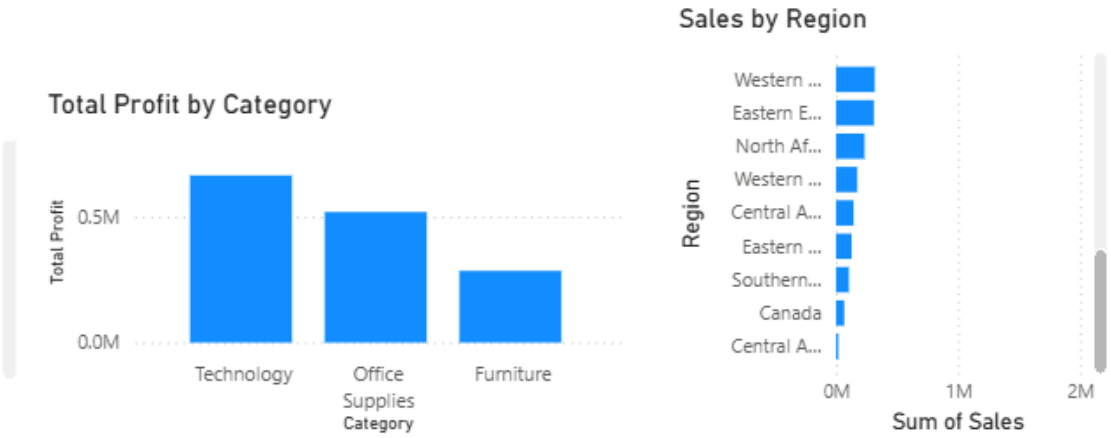
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Part D: Create Business Scorecard

1. Created a separate Business Scorecard page.
2. Added Sales Target vs Actual Sales.
3. Added Profit Target vs Actual Profit.
4. Added Order Target vs Actual Orders.
5. Applied conditional formatting to indicate performance status.
6. Used indicators such as Achieved, Partially Achieved, and Not Achieved.

Part E: Business Analysis

1. Analyzed dashboard findings.
2. Identified the best-performing and worst-performing regions.
3. Identified the most profitable and least profitable categories.
4. Prepared a business recommendation based on analytical findings.



The following KPI values were obtained from the Executive Dashboard:

KPI	**Observed Value**
Total Sales	12.64M
Total Profit	1.47M
Total Orders	26K
Total Margin	12%(approx)

1. Best Performing Region : Western Europe
2. Worst Performing Region : Central Asia

3. Most Profitable Category : Technology
4. Least Profitable Category : Furniture

11. RESULT

An interactive Executive Dashboard and Business Scorecard were successfully developed using Microsoft Power BI. Key business metrics were visualized through KPI cards, charts, and slicers, enabling effective performance monitoring and business analysis.

The Business Scorecard provides a comparison of actual performance against predefined targets, helping management evaluate organizational performance and make data-driven decisions.

12. LEARNING OUTCOMES

1. Design interactive dashboards using Power BI visualizations.
2. Create KPI-based scorecards for performance monitoring.
3. Apply visualization best practices in dashboard design.
4. Analyze business performance using dashboard insights.
5. Develop recommendations based on data-driven findings.